1. CREATE TABLES WITH CONSTRAINTS AND DEFINE TABLE STRUCTURE

```
postgres=# create database pl;
CREATE DATABASE
postgres=# \c pl
You are now connected to database "pl" as user "postgres".
pl=# create table student( stid int primary key, st_name text unique, marks int not null);
CREATE TABLE
```

SQL Shell (psql)

```
pl=# create table course( cid int primary key, c_name text, stid int references student(stid));
CREATE TABLE
pl=# \d student
                Table "public.student"
 Column  |  Type   | Collation | Nullable | Default
---------+---------+-----------+----------+---------
 stid    | integer |           | not null |
 st_name | text    |           |          |
 marks   | integer |           | not null |
Indexes:
    "student_pkey" PRIMARY KEY, btree (stid)
    "student_st_name_key" UNIQUE CONSTRAINT, btree (st_name)
Referenced by:
    TABLE "course" CONSTRAINT "course_stid_fkey" FOREIGN KEY (stid) REFERENCES student(stid)


pl=# \d course
                Table "public.course"
 Column |  Type   | Collation | Nullable | Default
--------+---------+-----------+----------+---------
 cid    | integer |           | not null |
 c_name | text    |           |          |
 stid   | integer |           |          |
Indexes:
    "course_pkey" PRIMARY KEY, btree (cid)
Foreign-key constraints:
    "course_stid_fkey" FOREIGN KEY (stid) REFERENCES student(stid)
```

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2. TO PERFROM JOINS WE NEED TO ADD DATA TO THE TABLES FIRST

SQL Shell (psql)

```
pl=# insert into student values (1,'carl',25),(2,'james',12),(3,'ken',45),(4,'john',89);
INSERT 0 4
pl=# insert into course values (101,'pl/sql',1),(111,'ICP',2),(121,'MIS',3),(131,'C',4);
INSERT 0 4
```

AFTER DATA INSERTION.....

SQL Shell (psql)

```
pl=# select * from student;
 stid | st_name | marks
------+---------+-------
    1 | carl    |    25
    2 | james   |    12
    3 | ken     |    45
    4 | john    |    89
(4 rows)


pl=# select * from course;
 cid | c_name | stid
-----+--------+------
 101 | pl/sql |    1
 111 | ICP    |    2
 121 | MIS    |    3
 131 | C      |    4
(4 rows)
```

LET'S IMPLEMENT THE JOINS!!

```
pl=# select * from student s inner join course c on s.stid=c.stid;
 stid | st_name | marks | cid | c_name | stid
------+---------+-------+-----+--------+------
    1 | carl    |    25 | 101 | pl/sql |    1
    2 | james   |    12 | 111 | ICP    |    2
    3 | ken     |    45 | 121 | MIS    |    3
    4 | john    |    89 | 131 | C      |    4
(4 rows)


pl=# select * from student s left join course c on s.stid=c.stid;
 stid | st_name | marks | cid | c_name | stid
------+---------+-------+-----+--------+------
    1 | carl    |    25 | 101 | pl/sql |    1
    2 | james   |    12 | 111 | ICP    |    2
    3 | ken     |    45 | 121 | MIS    |    3
    4 | john    |    89 | 131 | C      |    4
(4 rows)


pl=# select * from student s right join course c on s.stid=c.stid;
 stid | st_name | marks | cid | c_name | stid
------+---------+-------+-----+--------+------
    1 | carl    |    25 | 101 | pl/sql |    1
    2 | james   |    12 | 111 | ICP    |    2
    3 | ken     |    45 | 121 | MIS    |    3
    4 | john    |    89 | 131 | C      |    4
(4 rows)
```

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```
pl=# select * from student s full join course c on s.stid=c.stid;
 stid | st_name | marks | cid | c_name | stid
------+---------+-------+-----+--------+------
    1 | carl    |    25 | 101 | pl/sql |    1
    2 | james   |    12 | 111 | ICP    |    2
    3 | ken     |    45 | 121 | MIS    |    3
    4 | john    |    89 | 131 | C      |    4
(4 rows)
```

BUT WHAT WOULD HAPPEN IF WE CHANGED ONE ROW IN COURSE TABLE LET'S CHECK IT OUT!!!

```
SQL Shell (psql)
pl=# update course set stid=3 where cid=131;
UPDATE 1
pl=# select * from course;
 cid | c_name | stid
-----+--------+------
 101 | pl/sql |    1
 111 | ICP    |    2
 121 | MIS    |    3
 131 | C      |    3
(4 rows)


pl=#
```

LET'S IMPLEMENT THE JOINS AGAIN

```
SQL Shell (psql)
pl=# select * from student s inner join course c on s.stid=c.stid;
 stid | st_name | marks | cid | c_name | stid
------+---------+-------+-----+--------+------
    1 | carl    |    25 | 101 | pl/sql |    1
    2 | james   |    12 | 111 | ICP    |    2
    3 | ken     |    45 | 121 | MIS    |    3
    3 | ken     |    45 | 131 | C      |    3
(4 rows)


pl=# select * from student s left join course c on s.stid=c.stid;
 stid | st_name | marks | cid | c_name | stid
------+---------+-------+-----+--------+------
    1 | carl    |    25 | 101 | pl/sql |    1
    2 | james   |    12 | 111 | ICP    |    2
    3 | ken     |    45 | 121 | MIS    |    3
    3 | ken     |    45 | 131 | C      |    3
    4 | john    |    89 |     |        |
(5 rows)


pl=# select * from student s right join course c on s.stid=c.stid;
 stid | st_name | marks | cid | c_name | stid
------+---------+-------+-----+--------+------
    1 | carl    |    25 | 101 | pl/sql |    1
    2 | james   |    12 | 111 | ICP    |    2
    3 | ken     |    45 | 121 | MIS    |    3
    3 | ken     |    45 | 131 | C      |    3
(4 rows)
```

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3. CREATE INDEX ON STUDENT TABLE AND CHECK ITS EXISTENCE

```
SQL Shell (psql)
pl=# create index st_indx on student( stid, st_name);
CREATE INDEX
pl=# \d student
                Table "public.student"
 Column  |  Type   | Collation | Nullable | Default
---------+---------+-----------+----------+---------
 stid    | integer |           | not null |
 st_name | text    |           |          |
 marks   | integer |           | not null |
Indexes:
    "student_pkey" PRIMARY KEY, btree (stid)
    "st_indx" btree (stid, st_name)
    "student_st_name_key" UNIQUE CONSTRAINT, btree (st_name)
Referenced by:
    TABLE "course" CONSTRAINT "course_stid_fkey" FOREIGN KEY (stid) REFERENCES student(stid)
```

4. CREATE A VIEW ON BOTH OUR TABLES AND USE IT

```
SQL Shell (psql)
pl=# create view student_course as select s.st_name, c.c_name from student s, course c where s.stid=c.stid;
CREATE VIEW
pl=# select * from student_course;
 st_name | c_name
---------+--------
 carl    | pl/sql
 james   | ICP
 ken     | MIS
 ken     | C
(4 rows)
```

5. READ FURTHER EXPLANATIONS BY CHECKING THE README.md FILE INCLUDED WITH THIS PROJECT